Parametric Methods

Density Estimation

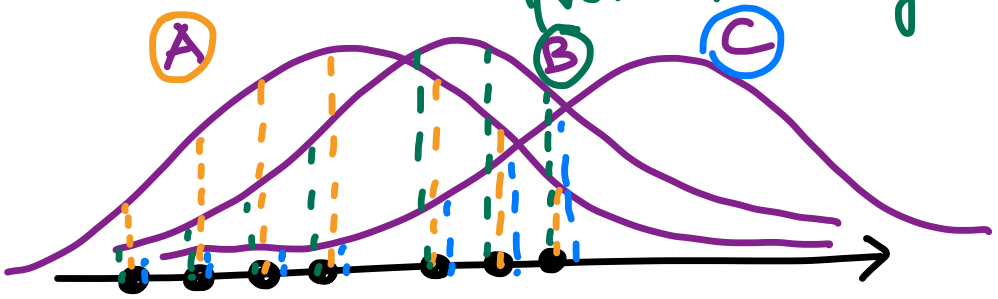
$X = \{x_i\}_{i=1}^{N}$ — N samples, N data points

$x_i \sim p(x_i) \quad \forall i \Rightarrow$ probability distribution

$\Downarrow$

parameters (?)

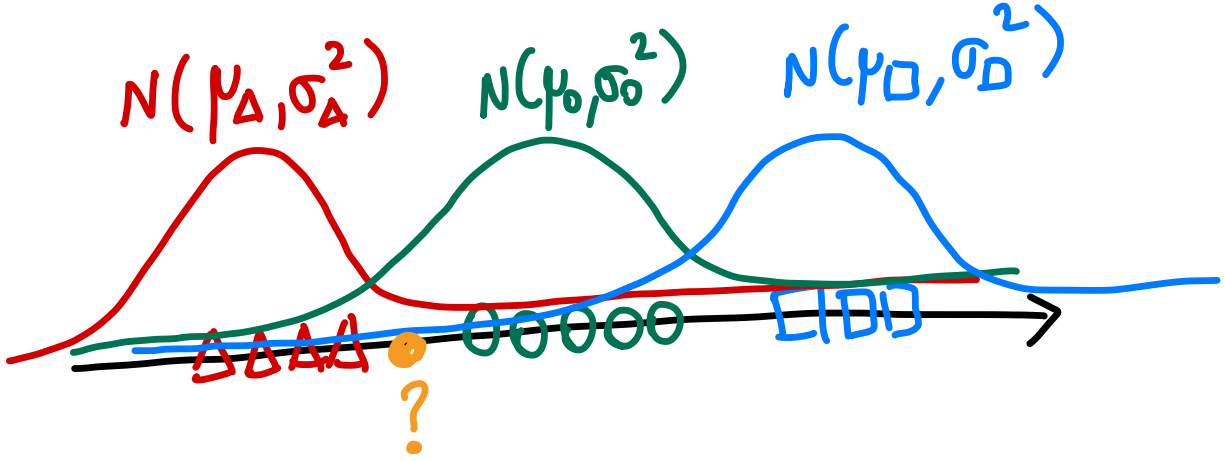
learn these parameters from training data



$x_i \sim N(x_i; \mu, \sigma^2)$

$\mu^* \Rightarrow$ best μ parameter

$\sigma^{2*} \Rightarrow$ best σ^2 parameter



$K = 3$ (# of classes)

$$\mathcal{X} = \{(x_i, y_i)\}_{i=1}^{N} \qquad x_i \in \mathbb{R}^1 \qquad y_i \in \{1, 2, 3\}$$

△ ○ □

density

$p(x \mid y=c) \Rightarrow$ class conditioned density

$Pr(y=c) \Rightarrow$ prior probability

probability

$$Pr(y=c \mid x) = \frac{p(x \mid y=c)\, Pr(y=c)}{p(x)}$$

A B

posterior probability

BAYES RULE $\Rightarrow Pr(A \mid B) = \dfrac{Pr(B \mid A)\, Pr(A)}{Pr(B)}$

predicted posterior probability

$$\hat{Pr}(y=c \mid x) = \frac{\hat{p}(x \mid y=c)\, \hat{Pr}(y=c)}{\hat{p}(x)}$$

x_{N+1} $\hat{Pr}(y=c \mid x_{N+1})$

- $\hat{Pr}(y=1 \mid x_{N+1})$ (△)
- $\hat{Pr}(y=2 \mid x_{N+1})$ (○)
- $\hat{Pr}(y=3 \mid x_{N+1})$ (□)

pick the one with the maximum probability

MAXIMUM LIKELIHOOD ESTIMATION (MLE)

$\chi = \{x_i\}_{i=1}^{N} \quad x_i \sim p(x_i|\theta) \quad \forall i$

↳ unknown parameters of p

x_i's are i.i.d. (identically & independently distributed)

$Pr(A,B) = Pr(A)Pr(B)$

$P(A,B) = P(A)\,P(B)$

$\text{Likelihood} \equiv p(x_1, x_2, \ldots, x_N|\theta) \Rightarrow$ full joint

$= p(x_1|\theta)\,p(x_2|\theta)\ldots p(x_N|\theta)$ ← i.i.d.

$\mathcal{L}(\theta) \equiv \prod_{i=1}^{N} p(x_i|\theta)$

$\log\text{likelihood} = \log\left[\prod_{i=1}^{N} p(x_i|\theta)\right]$ ← natural logarithm

$\log(a.b.c) = \log(a)+\log(b)+\log(c)$

$\log(a^b) = b.\log(a)$

$= \sum_{i=1}^{N} \log\left[p(x_i|\theta)\right]$

- Gaussian → exp
- Exponential → exp
- Bernoulli → power

Bernoulli Density

$0 < \pi < 1$ ↳ success probability

(H) success : π $\Rightarrow x = 1$

(T) failure : $1-\pi$ $\Rightarrow x = 0$

coin ⇒ H T H H H H T . . . . T

$x_1\ x_2\ x_3\ x_4\ x_5\ x_6\ x_7$ . . . . x_{100}

1 0 1 1 1 1 0 0

$1-x_i$

$$p(x_i|\pi) = \pi^{x_i}(1-\pi)^{1-x_i}$$

$$Pr(x_i = 1|\pi) = \pi^1 \cdot (1-\pi)^{1-1} = \pi \quad \text{success}$$

$$Pr(x_i = 0|\pi) = \pi^0 \cdot (1-\pi)^{1-0} = 1-\pi \quad \text{failure}$$

$$\mathcal{L}(\pi|\chi) = \prod_{i=1}^{N} \left[\pi^{x_i}(1-\pi)^{(1-x_i)}\right]$$

likelihood of your training set

$$\log\text{likelihood} = \sum_{i=1}^{N} \left[x_i \log(\pi) + (1-x_i)\log(1-\pi)\right]$$

$$\frac{\partial \log\text{likelihood}}{\partial \pi} = \sum_{i=1}^{N} \left[\frac{x_i}{\pi} + \frac{(1-x_i)\cdot(-1)}{1-\pi}\right] = 0 \Rightarrow \pi^* = \frac{\sum_{i=1}^{N} x_i}{N}$$

$\frac{\#\text{of heads}}{\#\text{of tosses}}$ Sample mean

$$\frac{\partial \log(a)}{\partial a} = \frac{1}{a}$$

$$\frac{\partial \log(1-a)}{\partial a} = -\frac{1}{1-a}$$

Gaussian Density

$$\mathcal{X} = \{x_i\}_{i=1}^{N}$$

$$x_i \sim N(x_i; \mu, \sigma^2) \Rightarrow \mu^* = ? \quad , \quad \sigma^{2*} = ?$$

$$\text{log likelihood} = \log \prod_{i=1}^{N} \left[\frac{1}{\sqrt{2\pi\sigma^2}} \cdot \exp\left[- \frac{(x_i - \mu)^2}{2\sigma^2} \right] \right]$$

$$x \sim N(x; \mu, \sigma^2)$$

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \cdot \exp\left[- \frac{(x-\mu)^2}{2\sigma^2} \right]$$

$$\text{log likelihood} = \sum_{i=1}^{N} \left[-\frac{1}{2} \cdot \log(2\pi\sigma^2) - \frac{(x_i - \mu)^2}{2\sigma^2} \right]$$

$$\frac{\partial \text{log likelihood}}{\partial \mu} = 0$$

$$\frac{\partial \text{log likelihood}}{\partial \sigma^2} = 0$$

Solve for μ^* and σ^{2*}.

Solution:

$$\mu^* = \frac{\sum_{i=1}^{N} x_i}{N}$$

sample mean

$$\sigma^{2*} = \frac{\sum_{i=1}^{N} (x_i - \mu^*)^2}{N}$$

sample variance

Parametric Classification

$\mathcal{X} = \{(x_i, y_i)\}_{i=1}^{N}$ ⇐ Input

Output ⇒ A classifier

$$\hat{y}_{N+1} = \arg\max_{c=1}^{K} g_c(x_{N+1})$$

↳ score function for class #c

$$\hat{Pr}(y=c \mid x) = \frac{\hat{p}(x \mid y=c)\,\hat{Pr}(y=c)}{\hat{p}(x)}$$

$\hat{p}(x)$ → independent of class label c

proportional to ←

$$\hat{Pr}(y=c \mid x) \;\alpha\; \hat{p}(x \mid y=c)\,\hat{Pr}(y=c)$$

equal upto a constant ←

$$\log \hat{Pr}(y=c \mid x) \overset{+}{=} \log\left[\hat{p}(x \mid y=c)\right] + \log\left[\hat{Pr}(y=c)\right]$$

$$g_c(x)$$

$-\log[\hat{p}(x)]$

$$g_c(x) = \log\left[\hat{p}(x \mid y=c)\right] + \log\left[\hat{Pr}(y=c)\right]$$

$\hat{p}(x \mid y=c) \sim N(x; \mu_c, \sigma_c^2)$; Frequency of class #c

$$= \log\left[\frac{1}{\sqrt{2\pi\sigma_c^2}} \cdot \exp\left[-\frac{(x-\mu_c)^2}{2\sigma_c^2}\right]\right] + \log\left[\hat{Pr}(y=c)\right]$$

? ? ? ?

$\mu_c^* = ?$ $\quad \sigma_c^{2*} = ?$ $\quad \hat{Pr}(y=c) = ?$

$$\mu_c^* = \frac{\left[\sum_{i=1}^{N} x_i \, 1(y_i=c)\right]}{\sum_{i=1}^{N} 1(y_i=c)}$$

$\Delta \Rightarrow 4$
$O \Rightarrow 5$
$\square \Rightarrow 3$

$$\sigma_c^{2*} = \frac{\left[\sum_{i=1}^{N} (x_i - \mu_c^*)^2 \, 1(y_i=c)\right]}{\sum_{i=1}^{N} 1(y_i=c)}$$

$$\hat{Pr}(y=c) = \frac{N_c}{N} = \frac{\sum_{i=1}^{N} 1(y_i=c)}{N}$$

one function

$$1(\cdot) = \begin{cases} 1 & \text{if TRUE} \\ 0 & \text{otherwise} \end{cases}$$